WORKED SOLUTION

STEP I 2019, Question 10

Mechanics • Projectile Motion

STEP I 2019, Question 10

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

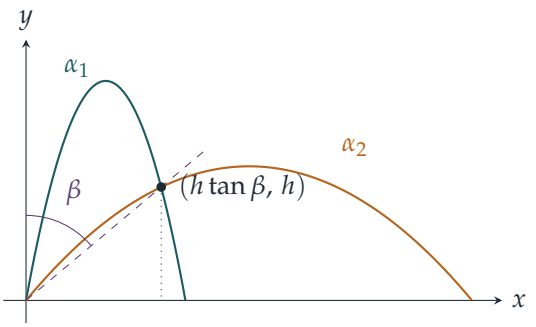
www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Write down the trajectory equation y in terms of x, put in the fixed point, and notice that $1/\sin^2\alpha = 1 + \cot^2\alpha$ turns it into a quadratic in $c = \cot\alpha$. After that, everything in (i) is the algebra of quadratics: sum and product of roots give $\alpha_1 + \alpha_2$ through the cot addition formula, and "two distinct roots" is a discriminant condition. Part (ii) is a one-line energy (or $v^2 = u^2 + 2as$) argument.

Solution

Model the particle as a point mass moving freely under gravity, with no air resistance. Take the angle α from the upward vertical, so the initial velocity has horizontal component $u\sin\alpha$ and vertical component $u\cos\alpha$.



The figure shows the two possible trajectories through the point (drawn for $\beta = 50°$, $k = 8$). The point lies on the line through O at angle β to the vertical.

Part (i)

At time t,

$$x = ut\sin\alpha, \qquad y = ut\cos\alpha - \tfrac{1}{2}gt^2.$$

Since $0 < \beta < 90^\circ$ the point has $x > 0$, so $\sin\alpha \neq 0$ and we may eliminate $t = x/(u\sin\alpha)$:

$$y = x\cot\alpha - \frac{gx^2}{2u^2\sin^2\alpha} = cx - \frac{gx^2}{2u^2}(1+c^2),$$

using $\csc^2\alpha = 1 + \cot^2\alpha$. Now put $x = h\tan\beta$, $y = h$ and divide by $h > 0$:

$$1 = c\tan\beta - \frac{gh}{2u^2}\tan^2\beta\,(1+c^2) = c\tan\beta - \frac{\tan^2\beta}{k}(1+c^2).$$

Multiplying through by $k\cot^2\beta$ gives $k\cot^2\beta = ck\cot\beta - 1 - c^2$, that is

$$c^2 - ck\cot\beta + 1 + k\cot^2\beta = 0, \qquad (*)$$

as required.

(a) The two values $c_1 = \cot\alpha_1$ and $c_2 = \cot\alpha_2$ are the two roots of the quadratic $(*)$ in c. (They are distinct because cot is one-to-one on $(0,\pi)$.) So, by the sum and product of roots,

$$\cot\alpha_1 + \cot\alpha_2 = k\cot\beta, \qquad \cot\alpha_1\cot\alpha_2 = 1 + k\cot^2\beta.$$

The first of these is the first result. For the second, use the addition formula for cot:

$$\cot(\alpha_1+\alpha_2) = \frac{\cot\alpha_1\cot\alpha_2 - 1}{\cot\alpha_1 + \cot\alpha_2} = \frac{k\cot^2\beta}{k\cot\beta} = \cot\beta.$$

? **Ask yourself** Does $\cot(\alpha_1+\alpha_2) = \cot\beta$ really force $\alpha_1 + \alpha_2 = \beta$?

Not on its own: cot has period 180°, so we also need to pin down the range of $\alpha_1 + \alpha_2$. Here both the sum $k\cot\beta$ and the product $1 + k\cot^2\beta$ are positive (as $k > 0$ and $\cot\beta > 0$), so both roots c_1, c_2 are positive. Hence both angles are acute, $0 < \alpha_1 + \alpha_2 < 180^\circ$, and on that interval cot is one-to-one. Since β is also in that interval, $\alpha_1 + \alpha_2 = \beta$.

(b) Two distinct real values of c require the discriminant of $(*)$ to be positive:

$$k^2\cot^2\beta - 4(1 + k\cot^2\beta) > 0.$$

Multiplying by $\tan^2\beta > 0$,

$$k^2 - 4k - 4\tan^2\beta > 0.$$

The quadratic in k on the left has roots $k = 2 \pm \sqrt{4 + 4\tan^2\beta} = 2 \pm 2\sec\beta$ (using $1 + \tan^2\beta = \sec^2\beta$ and $\sec\beta > 0$), so the inequality holds exactly when $k < 2 - 2\sec\beta$ or $k > 2 + 2\sec\beta$. But $\sec\beta > 1$, so $2 - 2\sec\beta < 0 < k$, and the first option is impossible. Therefore

$$k > 2(1 + \sec\beta).$$

Part (ii)

The particle must rise to height $h > 0$, so its initial vertical velocity $u\cos\alpha$ is positive. Vertically the motion has constant acceleration $-g$, and the vertical velocity is zero at the top, so by $v^2 = u^2 + 2as$ the greatest height is

$$H = \frac{u^2\cos^2\alpha}{2g}.$$

The particle passes through a point at height h, so $H \geq h$:

$$\frac{u^2\cos^2\alpha}{2g} \geq h \quad\Longrightarrow\quad \frac{2u^2}{gh}\cos^2\alpha \geq 4 \quad\Longrightarrow\quad k \geq 4\sec^2\alpha,$$

where we multiplied by $4/(gh) > 0$ and then divided by $\cos^2\alpha > 0$. Equality holds exactly when the point $(h\tan\beta, h)$ is the top of the trajectory.

Checks

- Take $\beta = 50^\circ$, $k = 8$. Then $2(1+\sec 50^\circ) \approx 5.11 < 8$, so two angles exist; solving $(*)$ gives $\alpha_1 \approx 10.29^\circ$ and $\alpha_2 \approx 39.71^\circ$, which add to 50°. Stepping the motion forward numerically, both trajectories reach height $1.0000\,h$ at $x = h\tan\beta$.
- For these angles $k\cos^2\alpha \approx 7.74$ and 4.73, both at least 4, as (ii) requires.
- A computer check over thousands of random β, h, k confirmed that $(*)$ has two distinct roots exactly when $k > 2(1+\sec\beta)$, and that $\alpha_1 + \alpha_2 = \beta$ every time.
- Units: $u^2/(gh)$ is dimensionless, as k must be.

Common slips

- Measuring α from the horizontal out of habit: here the horizontal component is $u\sin\alpha$, not $u\cos\alpha$.
- Concluding $\alpha_1 + \alpha_2 = \beta$ from $\cot(\alpha_1+\alpha_2) = \cot\beta$ without showing both angles are acute.
- Keeping the root $k < 2 - 2\sec\beta$ in (b), which is negative.

Beyond STEP

The result $\alpha_1 + \alpha_2 = \beta$ says the two firing directions are symmetric about the bisector of the vertical and the line to the target, a classical fact about the "parabola of safety": the envelope of all trajectories with speed u, which is the boundary case $k = 2(1+\sec\beta)$, where the two roots merge. Envelopes of families of curves are a nice first-year topic linking calculus and geometry. For more mechanics in this spirit, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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